

QUIZ PREPARATION

Organic Chemistry

Electrons → structure → bonding

Intuition · worked examples · retrieval practice

Prerequisites: the foundation

Before oxidation state, formal charge, or hybridization, you need four ideas:

1 · Valence electrons

The outer electrons available for bonding.

2 · Atomic orbitals

Regions where electrons are likely to be found.

3 · Lewis structures

Maps of bonds, lone pairs, and formal charges.

4 · Electron domains

Directions of electron density around an atom.

These ideas connect electron counting to molecular shape.

Valence electrons: the bonding budget

Valence electrons occupy the outermost shell and control most chemical bonding.

Atom	Valence e ⁻	Typical bonding pattern
H	1	one bond
C	4	four total bonds
N	5	three bonds + one lone pair
O	6	two bonds + two lone pairs
F / Cl	7	one bond + three lone pairs

Main-group shortcut: the periodic-table group tells you the valence-electron count.

Why atoms form bonds

Hydrogen seeks 2 electrons.

Its first shell contains only one $1s$ orbital, which can hold two electrons.

C, N, O, and F often seek 8.

Their valence shell contains one $2s$ and three $2p$ orbitals: four orbitals $\times$ two electrons.

A covalent bond lets two atoms share an electron pair. The shared pair counts toward the outer shell of **both** atoms.

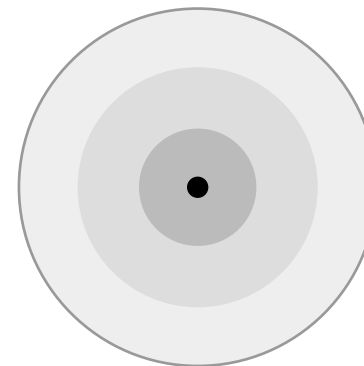
The octet rule is a useful pattern, not a universal law. Electron-deficient, odd-electron, and expanded-octet exceptions exist.

Orbitals: think “probability cloud”

An electron is not a tiny planet following a known circular path.

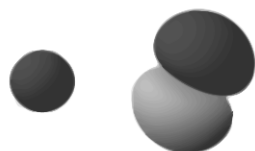
An **orbital** is a 3D probability pattern: a region where an electron is likely to be detected.

Analogy: a blurry long-exposure photo of a fast-moving fan shows where the blades are likely to be, not one exact blade position.



Darker center = greater probability near the nucleus. The cloud has no hard outer wall.

The shapes you need: s and p



1s 2p

s orbital: one sphere

Nondirectional; it looks the same from every angle.

p orbital: two lobes

The three *p* orbitals have the same shape but point along different axes.

The two shades mark opposite mathematical phases—not positive and negative electrical charges.

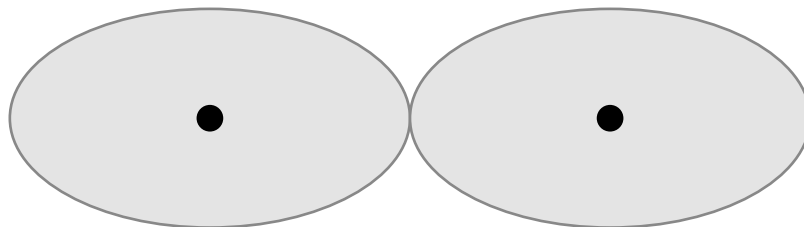
Orbital, subshell, and electron: don't mix them up

Word	Meaning	Capacity
orbital	one probability pattern	maximum 2 e ⁻
<i>s</i> subshell	contains one <i>s</i> orbital	maximum 2 e ⁻
<i>p</i> subshell	contains p_x , p_y , p_z	maximum 6 e ⁻

A “p orbital” is one dumbbell; the “p subshell” is the set of three dumbbells.

Why two electrons per orbital? They may share an orbital only when they have opposite spins.

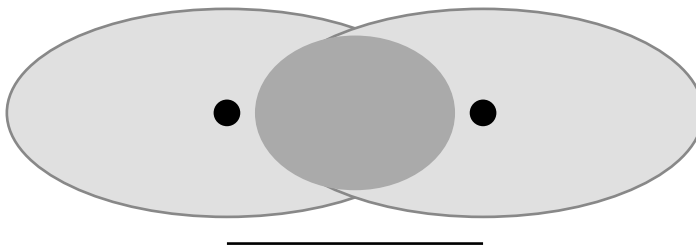
A bond forms where clouds overlap



Far apart, each nucleus attracts its own electron cloud. As the atoms approach, the clouds begin to overlap.

The useful question: where does shared electron density collect?

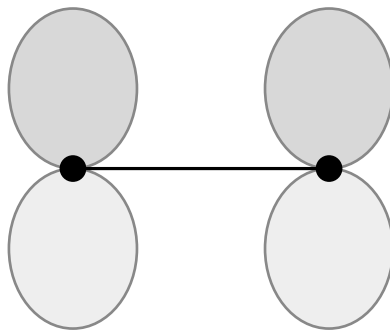
Head-on overlap makes a σ bond



The overlap sits directly **between the nuclei**, along the internuclear axis. Both nuclei attract this shared pair, holding the atoms together.

Answer Head-on overlap along the bond axis is a σ bond. Every single bond is one σ bond.

Side-by-side overlap makes a π bond



Parallel p orbitals touch sideways. Shared density appears **above and below** the bond axis, leaving a nodal plane through the nuclei.

Answer Side-by-side overlap is a π bond. It supplements—not replaces—the σ bond in a double bond.

Three overlap pictures at a glance



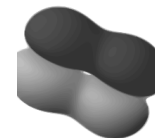
s-s σ

head-on spheres



p-p σ

head-on p lobes



p-p π

side-by-side p lobes

“ σ versus π ” describes the **symmetry of the overlap**, not whether the starting orbital was s, p, or hybrid.

Image adapted from Patricia.fidi, “Electron orbitals,” Wikimedia Commons, public domain.

Why hybrid orbitals are introduced

Hybridization is best understood as a **direction-making model**.

Raw atomic orbitals

Carbon starts with one spherical s and three perpendicular p orbitals.



Bond-pointing hybrids

Mix them into orbitals aimed toward neighboring atoms for stronger overlap.

4 directions

sp^3 , tetrahedral

3 directions

sp^2 , planar + one p left

2 directions

sp , linear + two p left

Hybrid orbitals form the σ framework; leftover p orbitals form π bonds.

What a Lewis structure shows

It shows

- which atoms are connected
- single, double, and triple bonds
- lone pairs
- formal charges

It does not directly show

- the molecule's 3D shape
- exact bond lengths
- true electron-density distribution
- electrons moving in fixed paths

one line = one shared electron pair = two electrons

Lewis vocabulary

Bonding pair

Two electrons shared between atoms.
Drawn as a line: $H - H$.

Lone pair

Two nonbonding electrons localized on one atom. Drawn as two dots.

Single / double / triple

One, two, or three shared pairs between the same atoms.

Formal charge

The charge produced when bonding electrons are divided equally.

Total electrons drawn must equal the available valence-electron total.

How to draw a Lewis structure

1. Count all valence electrons. Add electrons for a negative charge; subtract for a positive charge.
2. Choose a central atom. It is usually the least electronegative atom, never H.
3. Connect atoms with single bonds. Each line uses two electrons.

First build the skeleton; then distribute the remaining electrons.

How to draw a Lewis structure

1. Count all valence electrons.
2. Choose a central atom and draw single bonds.
3. Complete the terminal atoms' octets first.
4. Place any remaining electrons on the central atom.
5. If the center lacks an octet, convert neighboring lone pairs into multiple bonds.

How to draw a Lewis structure

1. Count electrons and construct the skeleton.
2. Complete octets; create multiple bonds if necessary.
3. Calculate every formal charge.
4. Verify that formal charges sum to the total molecular charge.
5. For equivalent arrangements, draw resonance structures.

Final audit: correct electron total, reasonable octets, and the smallest sensible formal charges.

Lewis example: carbon dioxide

Total valence electrons:

$$C : 4 + 2(O : 6) = 16 \text{ electrons}$$

Place carbon in the center and connect both oxygens:



Two single bonds use four electrons, leaving twelve to distribute.

Lewis example: carbon dioxide

Completing both oxygen octets uses all twelve remaining electrons.

$O - C - O$ with three lone pairs on each O

Each oxygen has an octet, but carbon has only four electrons around it.

Carbon still needs two more shared pairs.

Lewis example: carbon dioxide

Move one lone pair from each oxygen into a bonding pair:



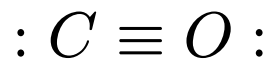
Now every atom has an octet, all 16 electrons are accounted for, and every formal charge is zero.

Answer The best Lewis structure contains two C=O double bonds.

Lewis example: carbon monoxide

CO has $4 + 6 = 10$ valence electrons.

A triple bond and one lone pair on each atom use all ten:



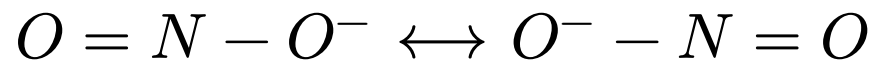
Carbon formal charge: $4 - 2 - 3 = -1$

Oxygen formal charge: $6 - 2 - 3 = +1$

The charges look unusual, but this arrangement gives both atoms complete octets.

Resonance: one structure may not be enough

Sometimes two or more valid Lewis structures differ only in electron placement—not atom positions.



The real molecule is a resonance hybrid: its electron density is delocalized, and the two N-O bonds are equivalent overall.

Resonance structures are not molecules flipping back and forth. They are multiple drawings of one delocalized electronic structure.

The one story behind this quiz

1 · Who gets electrons?

Electronegativity explains oxidation state and bond polarity.



2 · How do we count?

Lewis structures explain formal charge and hidden H atoms.



3 · What shape results?

Electron domains explain hybridization and orbital overlap.

Goal: learn a small set of electron-accounting ideas—not five unrelated rules.

Two bookkeeping systems

Oxidation state

Pretend every bond is **fully ionic**.

Give both bonding electrons to the more electronegative atom.

Useful for redox bookkeeping.

Formal charge

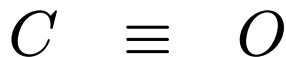
Pretend every bond is **perfectly covalent**.

Split bonding electrons equally between the atoms.

Useful for comparing Lewis structures.

Same molecule · different questions · different answers

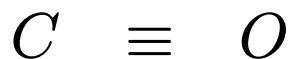
Oxidation state: follow the electron tug-of-war



Oxygen is more electronegative than carbon.

Which atom “wins” all six bonding electrons?

Oxidation state: follow the electron tug-of-war



Oxygen gets its usual -2 ; a neutral molecule must total zero:

$$x + (-2) = 0 \longrightarrow x = +2$$

Answer Carbon in CO has oxidation state $+2$.

Oxidation-state shortcuts—and why they work

O is usually -2

H is usually $+1$ with nonmetals

A neutral molecule sums to 0

An ion sums to its charge

Mental model

For each bond, ask: “If this bond broke heterolytically, which atom would keep the pair?”

These are **assigned** charges, not necessarily real partial charges.

Exceptions exist (peroxides, hydrides), but the quiz examples use the usual rules.

Worked set: carbon oxidation states

Molecule	Equation	Carbon state
CH ₄	$x + 4(+1) = 0$	-4 — C wins C-H electrons
CO	$x + (-2) = 0$	+2
CO ₂	$x + 2(-2) = 0$	+4
CH ₃ OH	$x + 4(+1) + (-2) = 0$	-2

Pattern: more C-O bonds make carbon more oxidized; more C-H bonds make it more reduced.

Formal charge: ownership under equal sharing

$$FC = V - N - \frac{B}{2}$$

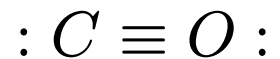
V: valence electrons of the isolated neutral atom

N: nonbonding electrons drawn on that atom

B/2: half of its bonding electrons

Fast version: valence – dots – bond lines

Formal charge of carbon in CO



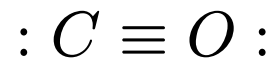
Count only what belongs to carbon:

Valence $V = 4$

Dots $N = ?$

Bond electrons $B = ?$

Formal charge of carbon in CO



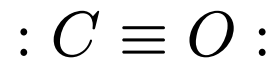
Valence $V = 4$

One lone pair: $N = 2$

Triple bond: $B = 6$

$$\text{FC} = 4 - 2 - \frac{6}{2}$$

Formal charge of carbon in CO



$$FC_C = 4 - 2 - 3 = -1$$

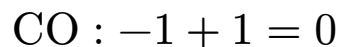
$$FC_O = 6 - 2 - 3 = +1$$

Answer CO is neutral because $(-1) + (+1) = 0$. Carbon: oxidation state +2, formal charge -1 .

Sanity checks for formal charge

Check 1 · Sum

All formal charges must sum to the molecule's total charge.



Check 2 · Plausibility

Prefer small formal charges and place negative charge on the more electronegative atom—when possible.

CO is an instructive exception to the preference: satisfying both octets forces C^- and O^+ .

WORKED EXAMPLE

Example: oxygen in carbon dioxide



For either oxygen: 6 valence electrons, 4 nonbonding electrons, 4 bonding electrons.

$$FC_O = 6 - 4 - \frac{4}{2} = 0$$

Answer Oxidation state = -2

Answer Formal charge = 0

Check yourself: carbon monoxide

What are the oxidation state and formal charge on carbon in CO, respectively?

A. 0; +1

B. +1; 0

C. +2; -1

D. +4; 0

Say your reason before advancing.

Check yourself: carbon monoxide

What are the oxidation state and formal charge on carbon in CO, respectively?

A. 0; +1

B. +1; 0

C. +2; -1

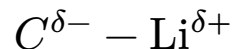
D. +4; 0

Answer Oxygen takes the bonding electrons in oxidation-state bookkeeping; equal sharing gives carbon formal charge -1.

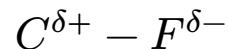
Electronegativity predicts bond polarity



Electron density shifts toward the more electronegative atom.



Carbon pulls harder than lithium.



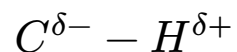
Fluorine pulls harder than carbon.

Partial charge is not formal charge

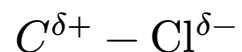
	Partial charge	Formal charge
Meaning	Unequal real electron density	Equal-share Lewis bookkeeping
Symbol	δ^+, δ^-	+1, 0, -1
Can be fractional?	Yes	No; integer in ordinary Lewis structures

A carbon can be δ^+ in a polar bond while having formal charge 0.

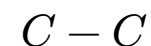
Polarity mini-set



Small polarity; carbon is slightly more electronegative.



Chlorine pulls electron density.



Same electronegativity; nonpolar bond.

Answer Among C-Li, C-Cl, C-C, and C-F, carbon is partially negative only in C-Li.

Skeletal structures are a compression code

What is shown

- A corner or unlabeled end = carbon
- A line = a bond
- Heteroatoms are written explicitly
- Hydrogens on heteroatoms are usually shown

What your brain supplies

- Carbon seeks total bond order 4
- Missing carbon bonds are C-H bonds
- Double lines count 2; triple lines count 3
- Charge can change the usual count

$$H_C = 4 - \text{bond order already drawn}$$

Count hidden hydrogens locally



Treat each carbon as a separate four-slot puzzle.

Left carbon: ? H

Middle carbon: ? H

Right carbon: ? H

Count hidden hydrogens locally



Count drawn bond order around each carbon.

Left: $4 - 1 = 3$ H

Middle: $4 - (1 + 2) = 1$ H

Right: $4 - 2 = 2$ H

Count hidden hydrogens locally



Total hidden hydrogen count: $3 + 1 + 2 = 6$

Answer The molecule is propene, C_3H_6 . The global formula confirms the local count.

Hydrogen-counting patterns worth recognizing

Carbon pattern	H	Reason
$-\text{CH}_3$	3	one bond order already used
$-\text{CH}_2-$	2	two single bonds used
$-\text{CH} =$	1	single + double = 3
$=\text{CH}_2$	2	one double bond = 2
internal alkyne C	0	triple + single = 4
substituted aromatic C	0	three bond-order units + substituent

A global cross-check: degree of unsaturation

For a formula containing C, H, N, and halogens X :

$$\text{DBE} = \frac{2C + 2 + N - H - X}{2}$$

One ring or one double bond contributes 1 DBE.

One triple bond contributes 2 DBE.

Example: C_6H_6 gives $\frac{12+2-6}{2} = 4$: one ring + three double bonds.

Hybridization: count directions, not bonds

A multiple bond points in **one** direction, so it counts as one electron domain.

Domains	Hybrid	Geometry	Angle
4	sp^3	tetrahedral	$\approx 109.5^\circ$
3	sp^2	trigonal planar	$\approx 120^\circ$
2	sp	linear	180°

Why “sp” labels encode orbital mixing



1 s + 3 p → 4 hybrid orbitals; no unhybridized p remains.



1 s + 2 p → 3 hybrids; one p remains for one π bond.



1 s + 1 p → 2 hybrids; two p orbitals remain for two π bonds.

This is why double bonds usually imply sp^2 and triple bonds imply sp .

Hybridization worked examples

CH_4 : four σ directions $\rightarrow sp^3$

$\text{CH}_2 = \text{CH}_2$: three directions at each C
 $\rightarrow sp^2$

$\text{HC} \equiv \text{CH}$: two directions at each C $\rightarrow sp$

Benzene: three directions at each C $\rightarrow sp^2$

Lone pairs count too: in an amine, N has three bonds + one lone pair $\rightarrow$ four domains $\rightarrow$ usually sp^3 .

σ and π bonds: two kinds of overlap

σ bond

Head-on overlap along the internuclear axis.

Every single bond is σ ; every multiple bond contains exactly one σ .

π bond

Side-by-side overlap of unhybridized p orbitals.

Double bond: 1 π . Triple bond: 2 π .



Name the orbital overlap

An indicated σ bond joins a benzene-ring carbon to an alkyne carbon.



First identify each endpoint independently.

Name the orbital overlap

benzene C – C $\equiv$ C

Benzene carbon: three domains $\rightarrow sp^2$

Alkyne carbon: two domains $\rightarrow sp$

Name the orbital overlap

benzene C – C $\equiv$ C

σ : $sp^2 - sp$ overlap

Answer The order of the labels does not matter: sp^2-sp is the same overlap as $sp-sp^2$.

Overlap examples

Bond	Endpoints	σ overlap
C-C in ethane	sp^3, sp^3	sp^3-sp^3
C=C in ethene	sp^2, sp^2	sp^2-sp^2
C $\equiv$ C in ethyne	sp, sp	$sp-sp$
C-H in ethene	$sp^2, 1s$	sp^2-1s
O-H in water	$sp^3, 1s$	sp^3-1s

Mixed practice: explain, don't guess

1. What are carbon's oxidation state and formal charge in CO?
2. Which carbon is δ^- : C-Li or C-F?
3. How many H atoms are attached to the middle carbon in $\text{CH}_3 - \text{CH} = \text{CH}_2$?
4. What is the hybridization of a carbon in a triple bond?
5. What orbitals form a σ bond between benzene C and alkyne C?

For each answer, name the rule that produced it. Retrieval + explanation builds durable memory.

Mixed practice: answers

1. CO carbon: oxidation state +2; formal charge -1 .
2. C-Li: carbon is δ^- because C is more electronegative.
3. Middle carbon has 1 H: $4 - (1 + 2) = 1$.
4. Triple-bond carbon is sp : two electron-domain directions.
5. The σ bond is sp^2-sp overlap.

Answer If any answer felt slow, revisit the matching “intuition” slide before memorizing a shortcut.

A reliable exam workflow

1. Draw or inspect the Lewis structure.

2. Count lone pairs and bond orders.

3. Decide which bookkeeping system is asked for.

4. Count local electron domains.

5. Label each atom's hybridization.

6. Check charge sums and carbon valence.

Final map

Electronegativity



oxidation state + bond polarity

Lewis electron counting



formal charge + hidden hydrogens

Electron-domain geometry



hybridization + σ overlap

Quiz answer key

Q	Answer	Key idea
1	C	CO carbon: +2 oxidation state, −1 formal charge
2	A	CO ₂ oxygen: −2 oxidation state, 0 formal charge
3	A	C is partially negative in C-Li
4	D	18 hidden hydrogens
5	C	sp^2 - sp σ overlap

Understand the reason—not only the letter.